

Report
Of
DATA ANALYTICS WITH PYTHON & POWER BI

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number of Student:

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Batch: 2024-28

2026-27

CANDIDATE'S DECLARATION

I, SHAGUN SRIVASTAVA, do hereby declare that the internship report titled “Data Analytics with Python & Power BI” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

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Student Name: SHAGUN SRIVASTAVA

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INTERNSHIP COMPLETION CERTIFICATE

CERTIFICATE
OF INTERNSHIP







PRESENTED TO

Shagun Srivastava

has successfully achieved the Internship Credential by completing a 8-week program in the domain of

Data Analytics with Python & Power BI

During the internship, the learner demonstrated proficiency in:

- Welcome to Data Analytics & Environment Setup & Core Python for Data Analysis
- Data Cleaning & Preprocessing with Pandas & SQL Fundamentals for Data Analysis
- Advanced SQL Techniques & Data Extraction & Data Visualization with Power BI - Basics
- Python & SQL Integration for Data Extraction & Power BI Data Modeling & DAX Basics
- Advanced Power BI Visualizations & Interactivity & Python for Data Visualization
- Python & Power BI Integration & Power BI Service & Collaboration
- Real-World Project: Sales Performance Analysis & Real-World Project: Customer Behavior Analysis & Best Practices & Project Workflow
- Capstone Project: E-commerce Sales & Customer Insights Dashboard

This certification recognizes the successful completion of the internship requirements.



Issued on: Aug 29, 2026
Certificate ID: 2026-CCCCC349D0



Mitu Swain
General Manager
(Learning & Development)
EduSkills

Certificate issued by EduSkills Academy for successful completion of the 8-week internship in Data Analytics with Python & Power BI.

4. PROJECT DESCRIPTION

4.1 Introduction

The internship was an 8-week program in the domain of Data Analytics with Python & Power BI. The program covered the practical workflow of data analytics, beginning with environment setup and core Python for data analysis and progressing through data cleaning, preprocessing, SQL, Power BI, DAX, visualization, integration, collaboration and project workflow.

The learning path connected Python-based data analysis with database querying and business intelligence. Pandas was used for data cleaning and preprocessing, SQL for data extraction and analytical querying, and Power BI for visualization, data modeling, interactivity and dashboard-oriented reporting. The program also included Python for data visualization and the integration of Python, SQL and Power BI.

The internship included real-world projects on Sales Performance Analysis and Customer Behavior Analysis and culminated in an E-commerce Sales & Customer Insights Dashboard. These components provided a project-oriented application of the technologies and concepts covered during the program.

4.2 Organization Profile

EduSkills Academy is the organization named on the internship completion credential. The supplied credential identifies the internship as an 8-week program in Data Analytics with Python & Power BI and lists the learning areas covered during the internship.

The documented training structure progressed from data analytics fundamentals and environment setup to Python, Pandas, SQL, Power BI, DAX, integration, real-world projects and a capstone dashboard. Since the supplied material contains no additional organizational profile, this report does not add unsupported information about the organization's history, workforce, clients or business operations.

4.3 Problem Statement

Business datasets can contain information about sales, customers and other activities, but raw data does not directly provide useful insights. It must be cleaned, transformed, queried, analyzed and visualized before patterns and trends can be communicated effectively.

The internship therefore focused on developing an end-to-end analytical workflow using Python, SQL and Power BI. For the capstone project, the specific focus was an

E-commerce Sales & Customer Insights Dashboard, with the objective of organizing sales and customer information into an interactive analytical view.

4.4 Project Objectives

- Understand the fundamentals of data analytics and establish an appropriate Python analysis environment.
- Develop practical proficiency in Python for data analysis.
- Perform data cleaning and preprocessing using Pandas.
- Use SQL fundamentals and advanced SQL techniques for data extraction.
- Develop data visualizations using Python and Power BI.
- Understand Power BI data modeling and the basics of DAX.
- Create advanced Power BI visualizations and interactive analytical views.
- Integrate Python with SQL for data extraction and connect Python with Power BI workflows.
- Apply analytical techniques to Sales Performance Analysis and Customer Behavior Analysis.
- Develop the capstone E-commerce Sales & Customer Insights Dashboard and follow best practices and project workflow.

4.5 Scope of the Project

The project covers the major stages of a practical data analytics workflow: understanding the data, cleaning and preprocessing, extracting relevant information, analyzing data, creating calculations, visualizing results and presenting insights through dashboards.

The internship scope includes Python, Pandas, SQL, Power BI, DAX, Python visualization, Power BI Service, Python and SQL integration, and Python and Power BI integration. The project scope further includes Sales Performance Analysis, Customer Behavior Analysis and the capstone E-commerce Sales & Customer Insights Dashboard.

The supplied internship credential does not provide the detailed dataset structure, numerical analysis results, source code, deployment configuration or dashboard screenshots. Such implementation-specific details have therefore not been invented in this report.

4.6 Technologies and Tools Used

S. No.	Technology / Tool	Purpose / Application
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1	Python	Core programming and data analysis
2	Pandas	Data cleaning, preprocessing and analysis
3	SQL	Data extraction and analytical querying
4	Power BI	Data visualization, dashboards and business intelligence
5	DAX	Power BI calculations and analytical measures
6	Python Visualization	Analytical data visualization
7	Power BI Service	Publishing, collaboration and service-based workflow
8	Python & SQL Integration	Combining Python analysis with SQL-based data extraction
9	Python & Power BI Integration	Connecting Python-based analysis with Power BI workflows

Table 4.1: Technologies and tools covered during the internship

4.7 System Architecture

The project workflow follows a layered data analytics architecture. Business data forms the input layer. The preparation layer cleans and preprocesses the data using Pandas. The extraction layer uses SQL to obtain required information. The analysis and modeling layer applies Python analysis together with Power BI modeling and DAX calculations. The visualization layer presents the results through interactive Power BI and Python visualizations. Power BI Service supports the final service and collaboration workflow.

Layer	Main Activity	Tools
1. Data Input	Access business data for analysis	Datasets / SQL
2. Data Preparation	Clean, transform and preprocess data	Python, Pandas
3. Data Extraction	Query and extract required information	SQL
4. Analysis & Modeling	Analyze data and create calculations	Python, Power BI, DAX
5. Visualization	Create interactive charts and dashboards	Power BI, Python
6. Integration & Collaboration	Connect tools and share analytical outputs	Python, Power BI Service

Figure 4.1: Logical system architecture of the internship project workflow

4.8 Methodology

The internship followed a progressive and project-oriented methodology in which each stage introduced analytical concepts and then connected them to practical project work.

Stage 1 – Data Analytics and Python Fundamentals: The program began with data analytics fundamentals, environment setup and core Python concepts required for data analysis.

Stage 2 – Data Cleaning and Preprocessing: Pandas was used for organizing, cleaning and preprocessing analytical data.

Stage 3 – SQL Fundamentals and Advanced SQL: SQL concepts progressed from fundamentals to advanced techniques for data extraction and analysis.

Stage 4 – Power BI Visualization: Power BI was introduced for basic visualization and then extended to advanced visualizations and interactivity.

Stage 5 – Data Modeling and DAX: Power BI data modeling and DAX basics were covered to support structured analytical calculations.

Stage 6 – Python, SQL and Power BI Integration: Python and SQL integration was covered for data extraction, while Python and Power BI integration connected programmatic analysis with dashboard workflows.

Stage 7 – Real-World Projects and Capstone: The learned skills were applied to Sales Performance Analysis and Customer Behavior Analysis and then to the E-commerce Sales & Customer Insights Dashboard.

Stage 8 – Power BI Service and Collaboration: Power BI Service and collaboration were included as part of the final analytics workflow.

4.9 Expected Outcomes

- Ability to use Python for practical data analysis.
- Improved understanding of data cleaning and preprocessing using Pandas.
- Ability to use SQL fundamentals and advanced techniques for data extraction.
- Understanding of Power BI visualization, interactivity and dashboard development.
- Basic understanding of Power BI data modeling and DAX.
- Ability to connect Python, SQL and Power BI in an analytical workflow.
- Practical exposure to sales performance and customer behavior analysis.
- Ability to structure an e-commerce sales and customer insights dashboard.
- Improved understanding of project workflow, best practices and Power BI Service collaboration.

4.10 Certificates of completion and other communication mail proof must be attached

The internship completion certificate is attached in this report as supporting proof of successful completion. The certificate identifies Shagun Srivastava as the learner and confirms completion of an 8-week Data Analytics with Python & Power BI program. It also lists the covered modules, real-world projects and the capstone E-commerce Sales & Customer Insights Dashboard.

No separate communication-mail proof was included in the supplied internship material. If required by the institute, the relevant official communication e-mails can be inserted after this section without changing the report format.

5. BIBLIOGRAPHY/REFERENCES

EduSkills Academy. Internship Completion Credential: Data Analytics with Python & Power BI. Issued August 29, 2026. Certificate ID: 2026-CCCCC349D0.
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